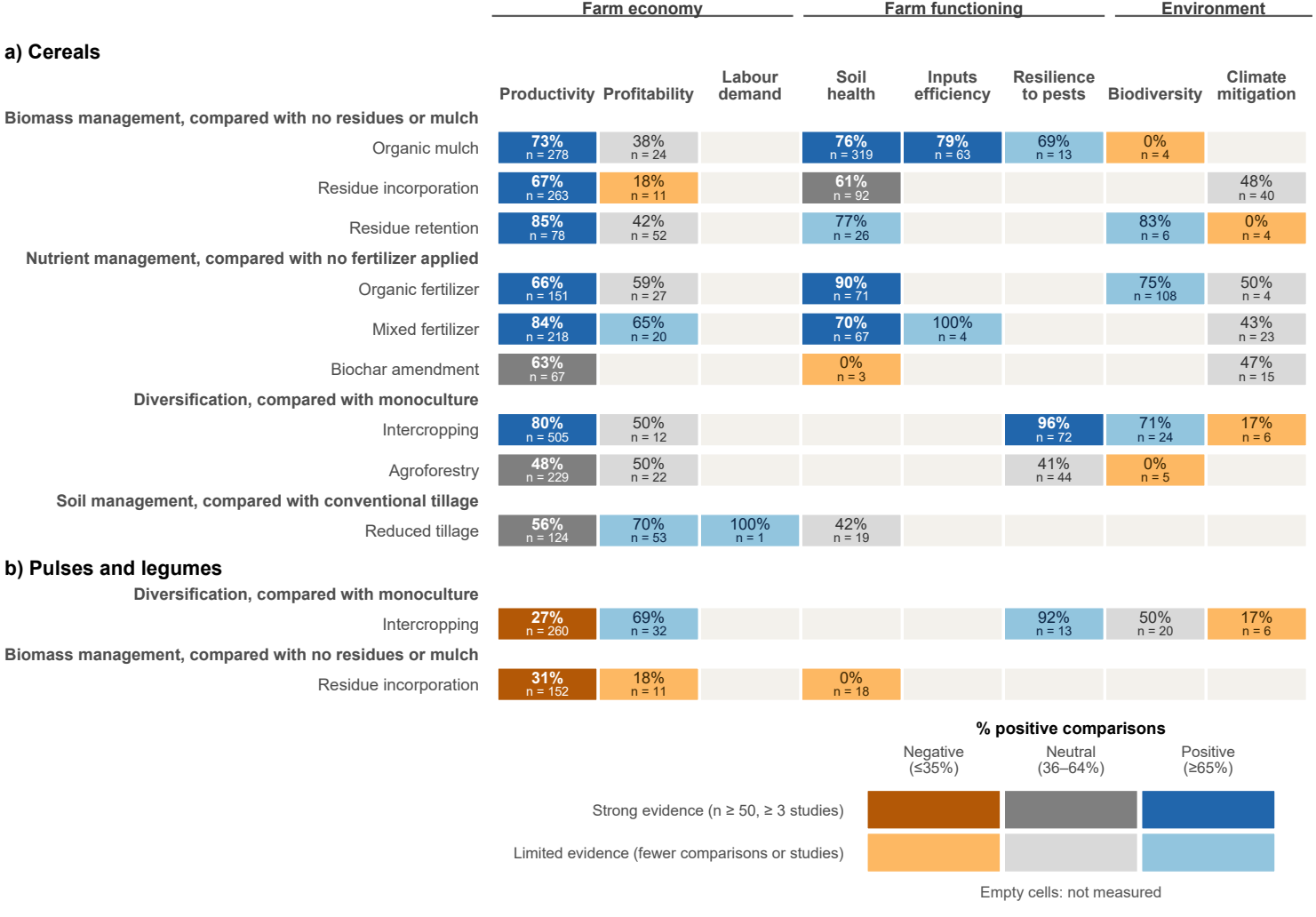
Farm economy
Farm functioning
Environment
a) Cereals
Productivity
Profitability
Labour demand
Soil health
Inputs efficiency
Resilience to pests
Biodiversity
Climate mitigation
Biomass management, compared with no residues or mulch
Organic mulch
73% n = 278
38% n = 24
76% n = 319
79% n = 63
69% n = 13
0% n = 4
Residue incorporation
67% n = 263
18% n = 11
61% n = 92
48% n = 40
Residue retention
85% n = 78
42% n = 52
77% n = 26
83% n = 6
0% n = 4
Nutrient management, compared with no fertilizer applied
Organic fertilizer
66% n = 151
59% n = 27
90% n = 71
75% n = 108
50% n = 4
Mixed fertilizer
84% n = 218
65% n = 20
70% n = 67
100% n = 4
43% n = 23
Biochar amendment
63% n = 67
0% n = 3
47% n = 15
Diversification, compared with monoculture
Intercropping
80% n = 505
50% n = 12
96% n = 72
71% n = 24
17% n = 6
Agroforestry
48% n = 229
50% n = 22
41% n = 44
0% n = 5
Soil management, compared with conventional tillage
Reduced tillage
56% n = 124
70% n = 53
100% n = 1
42% n = 19
b) Pulses and legumes
Diversification, compared with monoculture
Intercropping
27% n = 260
69% n = 32
92% n = 13
50% n = 20
17% n = 6
Biomass management, compared with no residues or mulch
Residue incorporation
31% n = 152
18% n = 11
0% n = 18
% positive comparisons
Negative (≤35%)
Neutral (36–64%)
Positive (≥65%)
Strong evidence (n ≥ 50, ≥ 3 studies)
Limited evidence (fewer comparisons or studies)
Empty cells: not measured